KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5421 | 5 Credits: 4 | Program core | Source-grounded

Unit Operations of Food Industry

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5421 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5421.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Food Processing Technology
Course code	5421
Course title	Unit Operations of Food Industry
Semester	5 Credits: 4
Course category	Program core
Periods per week	4(L:3 T:1 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	17 Understanding processing and different types of evaporators Describe the principles and mechanisms of freezing	See official syllabus
CO2	17 Understanding and freezing systems Describe the principles, working mechanisms, and applications of various food processing equipment,	See official syllabus
CO3	including hammer mills, ball mills, attrition mills, 13 Understanding and colloidal mills, in relation to energy laws and their effects on food. Describe the theory, importance, equipment, mechanisms, and applications of filtration,	See official syllabus
CO4	including plate and frame filters, rotary filters, and 13 Understanding the use of filter aids.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	of evaporators	3	As specified
Module 2	Describe the principles and mechanisms of freezing and freezing systems	4	As specified
Module 3	Describe the principles, working mechanisms, and applications of various	3	As specified
Module 4	applications of filtration, including plate and frame filters, rotary filters,	3	As specified

MODULE 1

# of evaporators

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of evaporators
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of

Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the principle of evaporation 6 understanding explain the working principle and mechanism of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect

7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding natural, forced circulation & falling thin film evaporator- describe single effect and multiple effect should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.

4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
of evaporators

  M1.01    Discuss the Principle of evaporation                 6
Understanding

           Explain the working principle and mechanism of
  M1.02                                                         7
Understanding
           Natural, forced circulation & falling thin film
           evaporator-
           Describe Single effect and multiple effect
  M1.03                                                         4
Understanding
           evaporator
Contents: Principles of evaporation, factors affecting rate of evaporation, types
of
evaporators, natural, forced, thin film evaporators etc. single effect, multiple
effect
evaporators.
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe the principles and mechanisms of freezing and freezing systems

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the principles and mechanisms of freezing and freezing systems
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe the Principle of freezing 7 Understanding List the factors influencing freezing time

Describe the Principle of freezing 7 Understanding List the factors influencing freezing time is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the Principle of freezing 7 Understanding List the factors influencing freezing time using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the principle of freezing 7 understanding list the factors influencing freezing time should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the Principle of freezing 7 Understanding List the factors influencing freezing time means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe the Principle of freezing 7 Understanding List the factors influencing freezing time □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Remembering Describe the direct freezing systems

3 Remembering Describe the direct freezing systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering Describe the direct freezing systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering describe the direct freezing systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering Describe the direct freezing systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Remembering Describe the direct freezing systems □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding Describe the Indirect freezing systems

3 Understanding Describe the Indirect freezing systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Describe the Indirect freezing systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding describe the indirect freezing systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Describe the Indirect freezing systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding Describe the Indirect freezing systems [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems

3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding factors influencing freezing time,freezing systems; direct contact systems(air blast, immersion) & indirect contact systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe the principles and mechanisms of freezing and freezing systems

  M2.01    Describe the Principle of freezing                   7
Understanding

           List the factors influencing freezing time
  M2.02                                                         3       Remembering
           Describe the direct freezing systems
  M2.03                                                         3
Understanding

           Describe the Indirect freezing systems
  M2.04                                                         3
Understanding
                             Series Test – I                    1
 Contents: Introduction, principles of freezing,. freezing point curve for food and
water,
 factors influencing freezing time,Freezing systems; Direct contact systems(air
blast,
 immersion ) & indirect contact systems
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe the principles, working mechanisms, and applications of various

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the principles, working mechanisms, and applications of various
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

laws 5 Understanding Discuss the equipment of size reduction in solid

laws 5 Understanding Discuss the equipment of size reduction in solid is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify laws 5 Understanding Discuss the equipment of size reduction in solid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, laws 5 understanding discuss the equipment of size reduction in solid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What laws 5 Understanding Discuss the equipment of size reduction in solid means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ laws 5 Understanding Discuss the equipment of size reduction in solid □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

food 5 Understanding Discuss the equipment of size reduction in liquid

food 5 Understanding Discuss the equipment of size reduction in liquid is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify food 5 Understanding Discuss the equipment of size reduction in liquid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, food 5 understanding discuss the equipment of size reduction in liquid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What food 5 Understanding Discuss the equipment of size reduction in liquid means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ food 5 Understanding Discuss the equipment of size reduction in liquid □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and

3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 food understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill describe the theory, importance, equipment, mechanisms, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 food Understanding mechanism and working principles of -– hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe the principles, working mechanisms, and applications of various
             food processing equipment, including hammer mills, ball mills,
attrition
             mills, and colloidal mills, in relation to energy laws and their
effects on
             food.
             State Principles of size reduction and Energy
   M3.01     laws                                                   5
Understanding
             Discuss the equipment of size reduction in solid
   M3.02     food                                                   5
Understanding

             Discuss the equipment of size reduction in liquid
   M3.03                                                            3
             food
Understanding

         Contents: Principles, types of equipments, energy laws, effect on foods,-
         mechanism and working principles of -– hammer mill, ball mill, attrition
mill,
         colloidal mill

             Describe the theory, importance, equipment, mechanisms, and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

applications of filtration, including plate and frame filters, rotary filters,

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: applications of filtration, including plate and frame filters, rotary filters,
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

importance of filtration

importance of filtration is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify importance of filtration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, importance of filtration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What importance of filtration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] importance of filtration [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss about the filtration equipment

Discuss about the filtration equipment is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss about the filtration equipment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss about the filtration equipment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss about the filtration equipment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Discuss about the filtration equipment [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html

Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize filter aids 4 understanding and diagrams with applications-plate and frame, rotary filters, filter aids-importance and applications text / reference t/r book title/author earle r l unit operations in food processing. web edition. pergamon press, r1 uk2004 fellows p j, food processing technology principles and practice, crc press, r2 2000. geankoplis c j, transport process and unit operations, prentice hall intl edition, r3 1993 albert ibarz and gustavo v. barbosa-cánovas, unit operations in food r4 engineering, crc press,2003 coulson, j.m. and etal. “coulson & richardson’s chemical engineering”, 6th r5 edition, vol. i & ii, butterworth – heinman / elsevier, 2004 foust, a.s. etal., “ principles of unit operations”, 2nd edition, john wiley & r6 sons, 1999. r. paul singh and dennis r. heldman, introduction to food engineering, 5th ed. r7 elsevier ,2014 online resources sl.no website link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3

https://savree.com/en/encyclopedia/ball-mill 4
https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html
□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
applications of filtration, including plate and frame filters, rotary filters,
             and the use of filter aids.
             Explain the theory of Filtration,
   M4.01     importance of filtration                               4
Understanding

   M4.02     Discuss about the filtration equipment                 4
Understanding

   M4.03     Summarize filter aids                                  4
Understanding

                               Series Test – II                     1
 Contents: Theory of filtration, importance of filtration, Filtration equipment,
mechanism
 and diagrams with applications-plate and frame, rotary filters, Filter aids-
importance and
 applications

Text / Reference

   T/R                                    Book Title/Author

         Earle R L Unit Operations in Food Processing. Web Edition. Pergamon
Press,
   R1    UK2004
           Fellows P J, Food Processing Technology Principles and Practice, CRC
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect. (3 marks)

Answer: Begin with the standard definition or identity of *7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Describe the Principle of freezing 7 Understanding List the factors influencing freezing time. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the Principle of freezing 7 Understanding List the factors influencing freezing time*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Remembering Describe the direct freezing systems. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering Describe the direct freezing systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Understanding Describe the Indirect freezing systems. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Describe the Indirect freezing systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain laws 5 Understanding Discuss the equipment of size reduction in solid. (3 marks)

Answer: Begin with the standard definition or identity of *laws 5 Understanding Discuss the equipment of size reduction in solid*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain food 5 Understanding Discuss the equipment of size reduction in liquid. (3 marks)

Answer: Begin with the standard definition or identity of *food 5 Understanding Discuss the equipment of size reduction in liquid*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and. (3 marks)

Answer: Begin with the standard definition or identity of *3 food Understanding mechanism and working principles of -– hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Module 4

Define or explain importance of filtration. (3 marks)

Answer: Begin with the standard definition or identity of *importance of filtration*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain Discuss about the filtration equipment. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss about the filtration equipment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000.

Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe the Principle of freezing 7 Understanding List the factors influencing freezing time**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **laws 5 Understanding Discuss the equipment of size reduction in solid**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **importance of filtration**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html
- https://www.sciencedirect.com/science/article/abs/pii/0032591075850042 https://savree.com/en/encyclopedia/ball-mill
- https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html

IN-PAGE SEARCH

Search this lesson